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Well Production Prediction Method Based on Multi-Factor Fusion Time Series Model

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Abstract

Production forecasting is a crucial issue in oilfield development, holding significant importance for evaluating geological potential and adjusting development schedule. However, due to complex geological conditions, seepage laws, and manual operations, traditional methods often struggle to achieve good results. This paper proposes a production forecasting method based on Multi-Factor LSTM combined with decline curve analysis. The production curve of an oil well is first fitted using a decline curve analysis, and then the difference between the actual production curve and the fitted production is taken as the input production data for the LSTM model. Considering operations such as daily production time, pump frequency changes, and choke size changes, this paper introduces these factors along with well production as input data within a time window to achieve higher production forecasting accuracy. The results indicate that the Multi-Factor LSTM combined with the decline curve analysis can effectively predict future well production. By introducing manual operations, the model can capture correlations between manual operations and oil production, demonstrating higher sensitivity to production fluctuations and achieving a model accuracy of 0.97. This represents an improvement of approximately 9% in R^2 compared to the Single-Factor LSTM model. Additionally, the introduction of the decline curve analysis complements the deficiency of machine learning methods in predicting long-term, monotonic decline trends. Compared to a standalone LSTM model, the proposed model better captures the long-term production decline due to natural energy depletion and reduced remaining oil reserves in the production sequence. The proposed oil well production forecasting method based on the time series model with multi-factor fusion exhibits good adaptability for oil well production forecasting, providing a new approach for rapid and accurate forecasting of field production and subsequent adjustment of production measures.

Introduction

Accurate production forecasting is crucial in the efficient development of oil and gas fields, involving evaluations of well production status and providing references for development schedule (Wen et al., 2023). Current production forecasting methods primarily include analytical methods, numerical calculation methods, empirical decline methods, and machine learning methods (Li et al., 2022; Wang et al., 2018; Zhang et al., 2022; Wang et al., 2020; Wang et al., 2023). Among them, analytical methods usually require coupling multiple seepage formulas, leading to complex and difficult calculation processes and stringent application conditions (Clarkson & Pedersen, 2010; Ozkan, 2001; Wang et al., 2018; Wu et al., 2016). Numerical calculation methods have a wide range of applications and high accuracy, but their prediction effectiveness depends on precise historical matching and high-quality geological modeling (Mukhina et al., 2021; Sun et al., 2016; Zhang et al., 2022; Zhou et al., 2019). Decline curve analysis tends to be overly smooth and ideal, suitable for long-term production forecasting (Arps, 1945; Fetkovich, 1973; Chen, 1979; Wang, 2019). However, the model does not consider the impact of actual production systems and other factors on well production, resulting in lower prediction accuracy for daily well production (Han et al., 2023; Wang et al., 2020; Wang et al., 2023).

In recent years, with the rapid development of machine learning methods, numerous scholars have recognized that oil well production is a classic time series problem and have begun to adopt different machine learning methods for oil and gas field production forecasting (Bagheri et al., 2020; Chahar et al., 2022; Huang et al., 2022). Single machine learning methods have been used for oil well production forecasting. For example, Xuanyi Song et al. optimized the hyperparameters of the LSTM neural network model using a particle swarm optimization algorithm and predicted the production of fractured horizontal wells (Song et al., 2020). Bagheri et al. employed the LSTM method to consider the impact of parameters such as temperature and pressure on oil well production (Bagheri et al., 2020). However, at this time, the impact of manual operations on production was not considered, and manually operated data, such as well shut-in operations, were deleted. Furthermore, single models suffer from issues such as local optima and poor computational accuracy, which are unable to meet the complexity of oil well production (Kumar et al., 2023; Li et al., 2022).

To improve model accuracy, scholars have introduced parameters such as manual operations and adopted machine learning methods for oil well production forecasting (Li et al., 2022; Liang et al., 2023; Ning et al., 2022; Wen et al., 2023). Xiangling Li et al. considered the impact of well shut-in/startup operations, choke size, and daily production time on oil well production (Li et al., 2022). Liu He et al. further explored the correlation between input data and oil well production, ranked the main controlling factors, conducted principal component analysis, and employed different machine learning methods to predict shale gas well production, providing guidance for the efficient development of shale gas wells at the site (Liu et al., 2023). The aforementioned methods merely mechanically analyze the impact of influencing factors on prediction accuracy, lacking practical physical significance (Wang et al., 2020; Wang et al., 2023; Zhang & Jia, 2021).

Some scholars have proposed algorithms coupling multiple factors and models (Mahdaviara et al., 2022; Ning et al., 2022; Sun et al., 2018). Wei Liu et al. proposed an EEMD-LSTM coupling model, using ensemble empirical mode decomposition to decompose residuals multiple times and improve model accuracy (Liu et al., 2020). Wei Bing et al. proposed an LSTM-SVR model considering three manual operation parameters—wellhead pressure, nozzle size, and flow rate—coupled with time series, enhancing model accuracy (Wen et al., 2023). Indrajeet Kumar introduced an attention mechanism into multi-factor machine learning methods in 2023, enabling adaptive weight allocation (Kumar et al., 2023). The aforementioned methods have gradually improved the accuracy of machine learning methods in oil well production forecasting. However, existing technologies are merely the coupling and superposition of mathematics and machine learning methods (Kumar et al., 2023; Liang et al., 2023; Ning et al., 2022; Wen et al., 2023), without considering the production decline laws of the reservoir itself. And this lacks physical significance in the oilfield applications.

Therefore, a new, fast, and effective production forecasting method should be established, one that can consider both the production decline caused by various formation and fluid factors and the fluctuating impact of manual operations on production. In response, this study proposes an oil well production forecasting model combining decline curve analysis and Multi-Factor LSTM fusion. It divides oil well production dynamics into linear and nonlinear categories, using the decline curve analysis to fit the natural production decline of oil wells and time series analysis to account for the impact of operations on oil well production, fully leveraging the advantages of decline curve analysis and machine learning methods.

Method

This paper conducts oil well production forecasting based on a time series model with multi-factor fusion, considering production decline due to geological factors and analyzing production fluctuations caused by manual operations in production data, combined with time series models for research. Depending on the composition of different production data, different methods are selectively used to forecast production data separately and merge the forecasting results. The data is divided into linear and nonlinear categories. Linear represents the natural decline of reservoir production over time, while nonlinear is due to production fluctuations caused by various factors. Therefore, the subsequent production forecasting process includes data preprocessing, decline curve analysis, Multi-Factor LSTM model analysis, and finally, decline curve analysis combined with Multi-Factor LSTM fusion production forecasting analysis.

Data Preprocessing

We selected production data and development dynamics from a single well in a certain block for analysis. This well has been in production since 2011, with a cumulative production time of 4,620 days up to 2024 and a cumulative production volume of 41.54×10^4 m^3. The sample data contains 17 parameters, which are categorized into four main groups: ① Production dynamics: daily liquid production, daily oil production, daily water production, and water cut; ② Manual operation parameters: daily production time, pump frequency, and choke size; ③ Pressure monitoring parameters: flowing pressure, oil pressure, pump inlet pressure, pump outlet pressure, backpressure, and casing pressure; ④ Temperature monitoring parameters: wellhead temperature, motor temperature, and flowing temperature.

Basic Statistical Description of Manual Operation Parameters' Original Data, as shown in the Table 1. Analysis suggests that the operation of shut-in/startup the well can lead to variations in daily production time, resulting in production fluctuations. Pump frequency determines the rate of fluid production. Increasing the pump frequency leads to an elevated daily fluid production rate, also causing production fluctuations. The size of the choke can alter the flow resistance of the oil well, thus forming production fluctuations. Manual operation parameters are the primary factors causing production fluctuations. Other parameters are dependent variables, representing parameter fluctuations caused by manual operations. Therefore, subsequent steps will involve selecting manual operation parameters for a multi-factor fused time series model for production forecasting.

Table 1—Basic Statistical Description of Manual Operation Parameters' Original Data

Parameter	Unit	Mean	Standard Deviation	Maximum	Minimum
Daily Production Time	h	22.81	5.12	24	0
Pump Frequency	Hz	41.37	11.24	50	0
Choke Size	mm	13.40	3.09	15.90	1.30

Decline Curve Analysis

In the initial stages of well production, the formation energy is abundant. As production continues, the oil production declines rapidly and then stabilizes at a lower level with minimal fluctuations as the crude oil is continuously extracted.

Common methods for decline curve analysis:

ARPS proposes that the decline rate has an exponential relationship with production and introduces the Arps model (Eq.1) for wells in the pseudo-steady flow stage with constant bottomhole flowing pressure (Fetkovich, 1973):

$$Q_t = \frac{Q_0}{\left(1 + D_i n t\right)^{1/n}} \qquad \text{Eq. 1}$$

VALKO considers cyclical characteristics and proposes the SEPD model (Eq.2), an extended exponential decline model suitable for cyclical linear flow stages (Valkó, 2009):

$$Q_t = Q_0 e^{-\left(\frac{t}{\tau}\right)^n} \qquad \text{Eq. 2}$$

Wang performs a Taylor expansion on traditional decline models, taking the first three terms to obtain a general formula and establish a combined model (Eq.3) (Wang et al., 2020):

$$Q_t = \frac{Q_0 + ct}{1 + at + bt^2} \qquad \text{Eq. 3}$$

Where: Q_t is the oil production at a certain time for the well, t/d; Q_0 is the initial or maximum oil production for the well, t/d; D_i is the decline rate, 1/d; t is the production time, d; m, n, τ, a, b, and c are formula coefficients, dimensionless.

Multi-Factor LSTM Nonlinear Forecasting

Oil well production forecasting is a typical time series forecasting problem related to the production history of oil and water wells. During production, engineering factors cause nonlinear fluctuations in production. Therefore, in the process of production forecasting, based on considering the temporal variation characteristics of production, the relationship between production indicators and manual operation parameters is also taken into account. The nonlinear variation patterns in production historical data during the development of oil wells are analyzed, and a production forecasting model is established using Multi-Factor LSTM.

The Multi-Factor LSTM production forecasting model considers the influence of multiple engineering factors on oil well production. The model consists of five layers: Input Layer, lstm Layer, dropout Layer, fully Connected Layer, and regression Layer.

Among them, the lstm Layer contains several LSTM nodes, which determine the intrinsic relationship between oil well production and engineering parameters through iterative calculations. The core of LSTM node design is the gating mechanism, including the input gate, forget gate, memory cell, and output gate, as shown in Fig. 1 (Li et al., 2022).

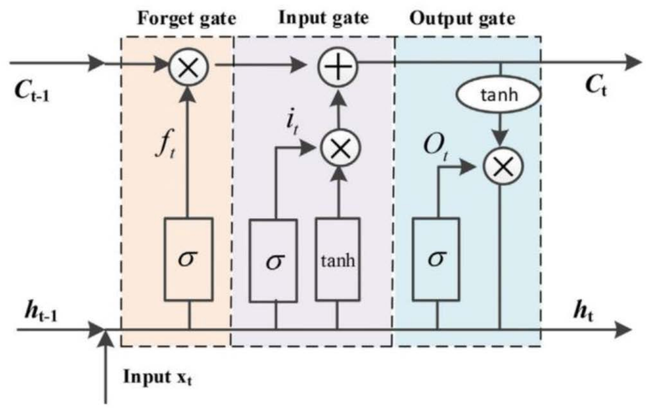

Figure 1—LSTM Node Structure (Li et al., 2022)

The input gate screens new information, determining which information to input into the memory cell state and updating the memory cell state at the current moment. It decides which memory information from the previous moment and which information should be retained, and multiplies them with the current input. The mathematical model (Eq.4-5) of the input gate is (Huang et al., 2022):

$$i_t = \sigma\left(W_i \cdot \left[h_{t-1}, x_t\right] + b_i\right) \quad \text{Eq. 4}$$

$$\widetilde{c}_t = \tanh\left(W_c \cdot \left[h_{t-1}, x_t\right] + b_c\right) \quad \text{Eq. 5}$$

The forget gate controls the influence of the memory from the previous time step on the current time step, deciding which information should be discarded. Its mathematical model (Eq.6) is (Song et al., 2020):

$$f_t = \sigma\left(W_f \cdot \left[h_{t-1}, c_t\right] + b_f\right) \quad \text{Eq. 6}$$

The memory cell state, similar to a conveyor belt, runs through the entire LSTM network, storing the network information of the current moment's LSTM and transmitting it downward. The mathematical model (Eq.7) for the memory cell state is (Liu et al., 2020):

$$c_t = f_t * c_{t-1} + i_t * \widetilde{c}_t \quad \text{Eq. 7}$$

The output gate determines the final output and retained information of the LSTM model. It uses the Sigmoid function to weight and sum the input information at the current moment and the output information from the previous moment, obtaining an initial output. Then, the previously learned memory information is scaled by the tanh activation function and multiplied by the initial output to obtain the final output of the LSTM model. The mathematical model (Eq.8-9) of the output gate is (Li et al., 2022):

$$o_t = \sigma\left(W_o \cdot \left[h_{t-1}, x_t\right] + b_o\right) \quad \text{Eq. 8}$$

$$h_t = o_t * \tanh\left(c_t\right) \quad \text{Eq. 9}$$

Where, W_f is the weight matrix of the forget gate, W_i is the weight matrix of the input gate, W_o is the weight matrix of the output gate, $[h_{t-1}, c_t]$ concatenates two vectors, b_f is the bias term of the forget gate, b_i is the bias term of the input gate, b_o is the bias term of the output gate, and σ is the sigmoid function.

To consider time correlation, the data is divided into input sets and corresponding output sets through time window segmentation. Using a time step n as the interval, the data from the first n time steps at each moment is taken as input, and the corresponding sample value at that moment is taken as the target output, resulting in a series of input-output pairs. To ensure that the input data contains multiple factors, the three main controlling factors of production and the historical production of n-1 time steps are integrated as input sets.

The output value at time t is the result of the combined effect of the input value of the current network unit and the output data of the unit at time t-1, indicating that the LSTM model can retain information from the previous time step and apply it to the current time step, demonstrating LSTM's learning ability on temporal data.

The model parameters are updated using the Adam algorithm through iterative optimization until the loss function reaches a convergent state. After model training, the output layer processes the results, such as denormalization, to restore the normalized predicted values to their original time series data format matching the actual production data, thereby completing the production forecasting.

This Multi-Factor LSTM-based production forecasting model considers the intrinsic relationships between oil production and geological as well as engineering factors, improving the accuracy and reliability of production forecasting.

Decline Curve Analysis and Multi-Factor LSTM Fusion

To fully leverage the advantages of both decline curve analysis and machine learning methods, different approaches are adopted to forecast production data for various compositions of production data, and the forecasting results are combined. Firstly, the decline curve analysis is used to fit the linear pattern of crude oil production decline due to continuous exploitation. Secondly, a multi-factor fusion machine learning method is employed to capture the nonlinear production fluctuations caused by human factors. Finally, a production forecasting method integrating decline curve analysis and Multi-Factor LSTM is constructed to enhance calculation accuracy. The calculation process is illustrated in the Fig. 2.

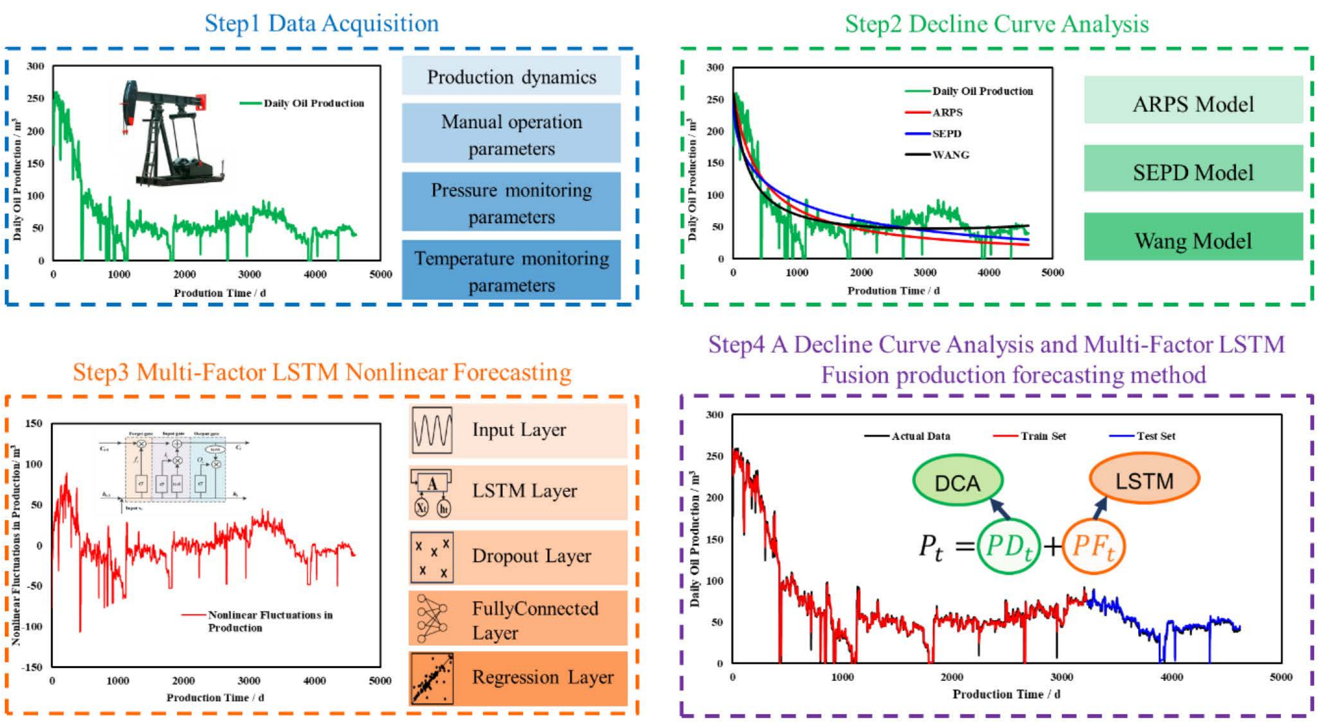

Figure 2—Decline Curve Analysis and Multi-Factor LSTM Fusion

On the basis of fitting conventional decline models, machine learning is incorporated to predict fluctuating data. The main processes include: Data Preprocessing, Decline Curve Analysis, Multi-Factor LSTM Nonlinear Forecasting, forming a Decline Curve Analysis and Multi-Factor LSTM Fusion production forecasting method.

Evaluation Metrics

To assess the accuracy and generalization ability of the model, the following evaluation metrics are used to measure the prediction performance of the model (Ning et al., 2022):

A. Mean Squared Error (MSE)

$$MSE = \frac{\sum_{i=1}^{N}\left(Q_i - \hat{Q}_i\right)^2}{N} \quad \text{Eq. 10}$$

B. Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N}\left(Q_i - \hat{Q}_i\right)^2}{N}} \quad \text{Eq. 11}$$

C. Mean Absolute Error (MAE)

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|Q_i - \hat{Q}_i\right| \quad \text{Eq. 12}$$

D. Coefficient of Determination (R^2)

$$R^2 = \frac{\sum_{i=1}^{N}\left(Q_i - \overline{Q}\right)^2}{\sum_{i=1}^{N}\left(\hat{Q}_i - Q_i\right)^2} \quad \text{Eq. 13}$$

Where N represents the data length, Q_i represents the original data, $\widehat{Q}_i$ represents the predicted results, and $\overline{Q}$ represents the average of the original data.

Results

Decline Curve Analysis

This paper uses daily oil production data from a single well that has been in production since 2011, with a cumulative production time of 4,620 days up to 2024. A time series plot is created as shown in the Fig. 3. Production declines rapidly in the early stage; after 1,000 days of production, daily production gradually flattens out at about 50 m^3/d. The entire production process follows the production decline law, but due to various factors, data fluctuations are apparent.

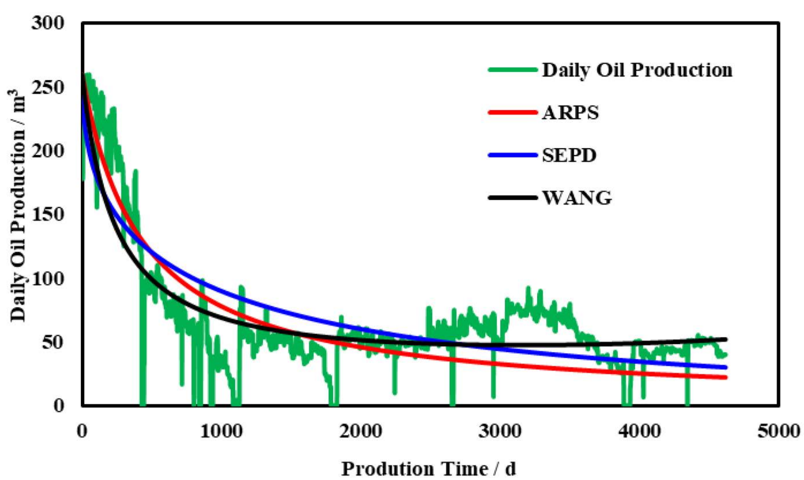

Figure 3—Daily Oil Production

Using the decline curve analysis for fitting, the R^2 values for the production decline fitting results of ARPS, SEPD, and WANG are: 0.5791, 0.5766, and 0.6436, respectively. The WANG fitting result is the best, as shown in the figure. The equation expression (Eq.14) after WANG fitting is:

$$Q_t = \frac{Q_0 + 0.05722 \times t}{1 + 0.00406 \times t - 4.513 \times 10^{-7} \times t^2} \quad \text{Eq. 14}$$

The daily oil production calculated using the decline curve analysis proposed by WANG is subtracted from the original daily oil production to obtain the nonlinear fluctuations production value caused by other factors, as shown in Fig. 4.

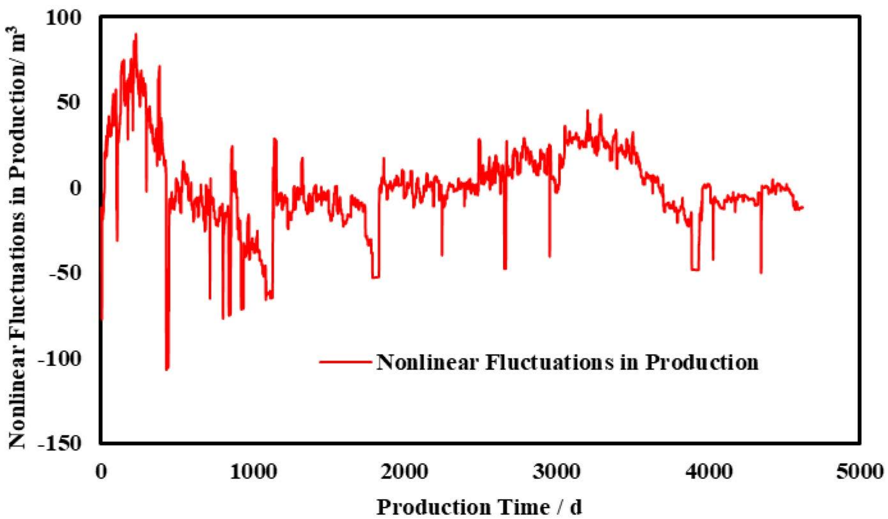

Figure 4—Nonlinear Fluctuations in Production

Multi-Factor LSTM Nonlinear Prediction

Due to production fluctuations caused by multiple factors, a time series method is used for production forecasting. For nonlinear fluctuating production data, the following three methods are used for comparative analysis: Single-Factor LSTM, Multi-Factor LSTM, and Multi-Factor GRU. The data is divided into two parts in a 7:3 ratio, with 70% of the data used as the training set for model training and the remaining 30% used as the test set to evaluate the accuracy of the model. The model utilizes the manually operated parameters and oil well production data from the first seven days to predict the oil well production for the eighth day.

The Fig. 5 shows the evaluation results of the R^2 for the training and test sets of different time series models. It can be found that the evaluation coefficients of the three methods in the training set are all above 0.9; and the multi-factor time series method is superior to the single-factor method; among them, the Multi-Factor LSTM has the best result in the training set, with a coefficient of determination of 0.97. The LSTM model is superior to GRU mainly due to its stronger long-term memory ability, higher model complexity, and ability to mitigate gradient vanishing, making it suitable for tasks requiring long-term dependencies and complex sequence analysis (Liang et al., 2023). The Multi-Factor LSTM is superior to the Single-Factor LSTM mainly because there are many factors influencing production fluctuations. Due to the unicity of its input parameters, the Single-Factor LSTM model neglects the impact of other significant factors on production fluctuations and simplifies its model structure, resulting in biased prediction outcomes. After inputting multiple parameters, the model can more comprehensively and systematically capture the correlations between data, increasing the prediction accuracy of the model (Kumar et al., 2023).

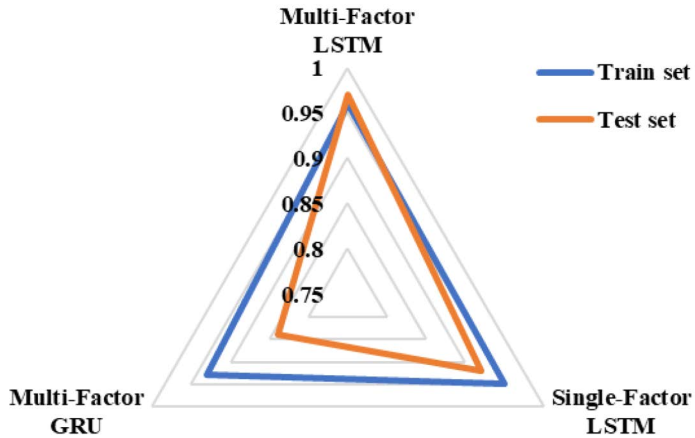

Figure 5—The R^2 of Different Time Series Models

As shown in the figure, the evaluation results of the training set and test set are not completely correlated. The evaluation results of the test set from high to low, are: Multi-Factor LSTM > Single-Factor LSTM> Multi-Factor GRU. The results of Multi-Factor LSTM production prediction are better than those of Single-Factor LSTM, mainly because the input parameters of single-factor are fewer, ignoring the influence of manual operation parameters on the prediction results, leading to poor prediction accuracy. Other evaluation parameters of the test set are shown in the Table 2, consistent with the results obtained by R^2. The other parameters of the Single-Factor LSTM are all higher than those of the Multi-Factor LSTM, approximately 1.5 times those of the Multi-Factor LSTM.

Table 2—Table Evaluation Metrics for Different Methods

Method	MSE	RMSE	MAE	R^2
Multi-Factor LSTM	6.93	2.67	1.36	0.96
Single-Factor LSTM	10.61	3.26	2.27	0.94
Multi-Factor GRU	31.83	5.64	3.79	0.87

Fusion Production Forecasting Results

Multiple methods such as Single-Factor LSTM, Multi-Factor LSTM, decline curve analysis combined with Single-Factor LSTM, and decline curve analysis combined with Multi-Factor LSTM are used for reservoir production forecasting models. The prediction results of each model are compared using untrained test set data, as shown in Table 3.

Table 3—Table Evaluation Metrics Results for Different Models

Method	MSE	RMSE	MAE	R^2
Single-Factor LSTM	34.27	5.85	4.24	0.89
Multi-Factor LSTM	22.95	4.79	2.97	0.92
Decline Curve Analysis + Single-Factor LSTM	10.72	3.27	2.29	0.95
Decline Curve Analysis + Multi-Factor LSTM	7.16	2.68	1.36	0.97

All evaluation metric results in Table 3 show that the production forecasting performance is: Single-Factor LSTM < Multi-Factor LSTM < Decline Curve Analysis + Single-Factor LSTM < Decline Curve Analysis + Multi-Factor LSTM. Through comparison, it is found that the decline curve analysis combined with LSTM production forecasting model performs better than models that rely solely on LSTM for production forecasting, with an improvement in R^2 by approximately 0.05. Multi-factor production forecasting performs better than single-factor forecasting, with an increase in R^2 by about 0.03. Among them, the decline curve analysis combined with Multi-Factor LSTM production forecasting method has the lowest RMSE, with an R^2 value of 0.97, which is closer to 1 and indicates the smallest prediction error. This represents an improvement of approximately 9% in R^2 compared to the Single-Factor LSTM model. This indicates that this model has better prediction performance and generalization ability than other methods. Analysis indicates that the introduction of multiple parameters allows the model to better capture the correlations between them, thereby increasing the accuracy of the prediction model.

Furthermore, incorporating decline curve analysis of production eliminates the linear impact of natural decline in reservoir production capacity, enhances the model's sensitivity to fluctuating data relationships, and thus improves model accuracy. Declining analysis combined with a Multi-Factor LSTM emerges as the optimal method for production capacity forecasting. The production forecasting method of decline curve analysis and Multi-Factor LSTM fusion is the optimal method. The results for both the training set and the testing set are illustrated as shown in Fig. 6. As shown in the Fig.6, this method recognizes that the oil well production is zero during shut-in operations.

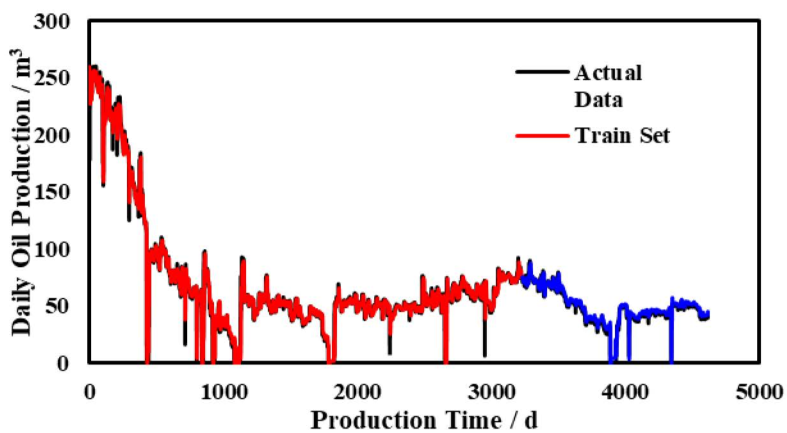

Figure 6—The Production Forecasting Results of Decline Curve Analysis and Multi-Factor LSTM Fusion

The deviations between the predicted and actual values for different methods are shown in the Fig. 7. It is observed that when the actual production is 0 m^3/d, during well shut-in operations, the deviation between the predicted and actual values is the largest. Comparing Fig.7 (a) with Fig.7 (b), it is found that when the actual production is 0 m^3/d, the fluctuation range of Single-Factor LSTM is larger than that of Multi-Factor LSTM. This indicates that during Single-Factor LSTM production prediction analysis, fewer constraint conditions are considered, leading to a reduction in model complexity and poorer accuracy and reliability of production prediction results, making it difficult to fully reflect the complexity and variability of actual conditions. Comparing Fig.7 (a~b) with Fig.7 (c~d), it is found that when production is 0 m^3/d, both LSTM methods without decline curve analysis produce negative values, deviating significantly from the actual values. The two methods combining decline curve analysis with LSTM for production capacity prediction perform significantly better in this scenario than the other two methods.

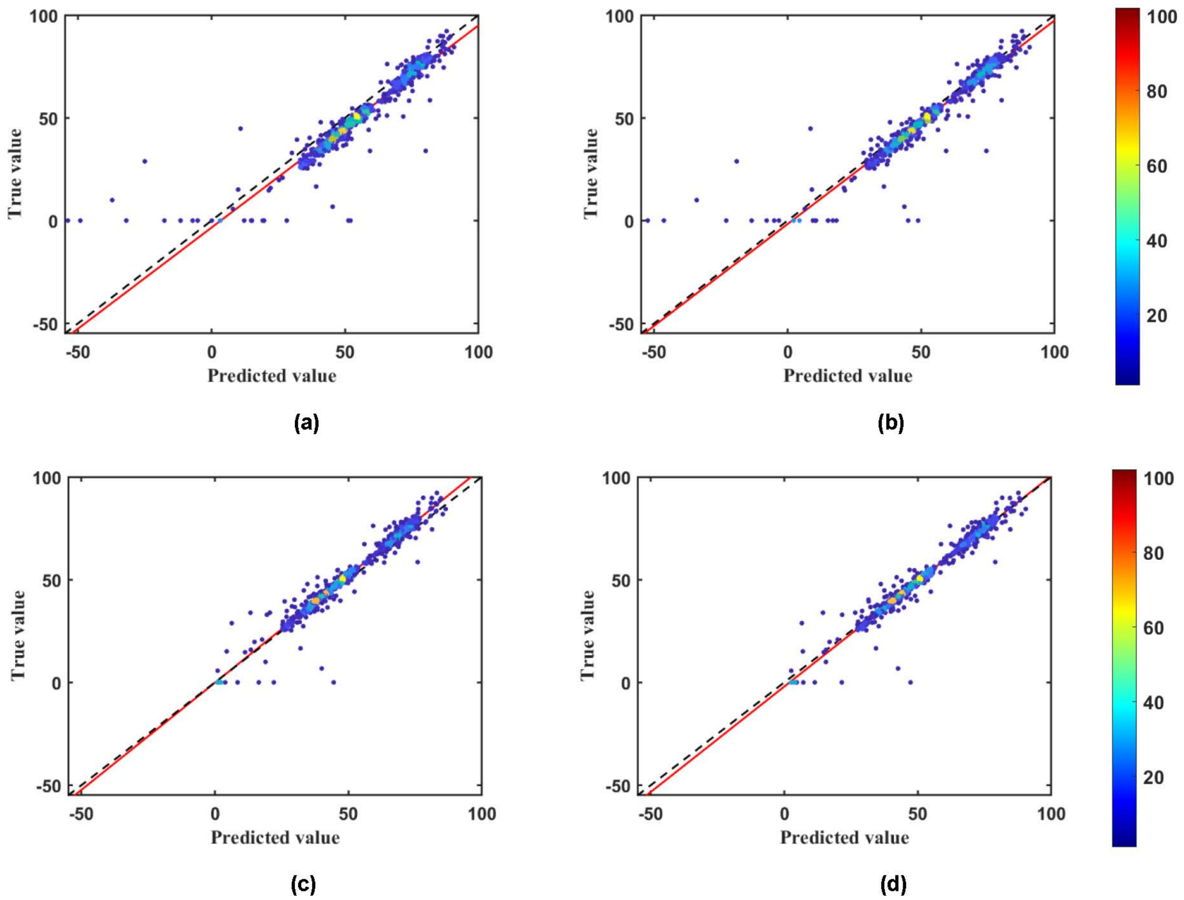

Figure 7—The Deviations Between the Predicted and Actual Values for Different Methods. (a) Single-Factor LSTM; (b) Multi-Factor LSTM; (c) Decline Curve Analysis + Single-Factor LSTM; (d) Decline Curve Analysis + Multi-Factor LSTM.

Using the same method to analyze the production data of another oil well, which commenced production in 2013 and had accumulated 3800 days of production by 2024, with a total cumulative oil production of 36.24×10^4 m^3, we again applied the ARPS, SEPD, and WANG decline curve analysis formulas for fitting. The R^2 values were 0.7704, 0.7813, and 0.7859, respectively. Finally, adopting the WANG decline curve analysis formula, the fitted equation can be expressed as Eq.15:

$$Q_t = \frac{Q_0 + 51.22\times t}{1 + 0.2788\times t + 2.93\times 10^{-4}\times t^2} \qquad \text{Eq. 15}$$

Nonlinear fluctuations in production are caused by multiple factors. And the above-mentioned multi-factor LSTM time series method is used for production forecasting. As shown in the Fig.8, the results for both the training set and the test set align well with the actual data, with final R^2 values of 0.947 and 0.968, respectively.

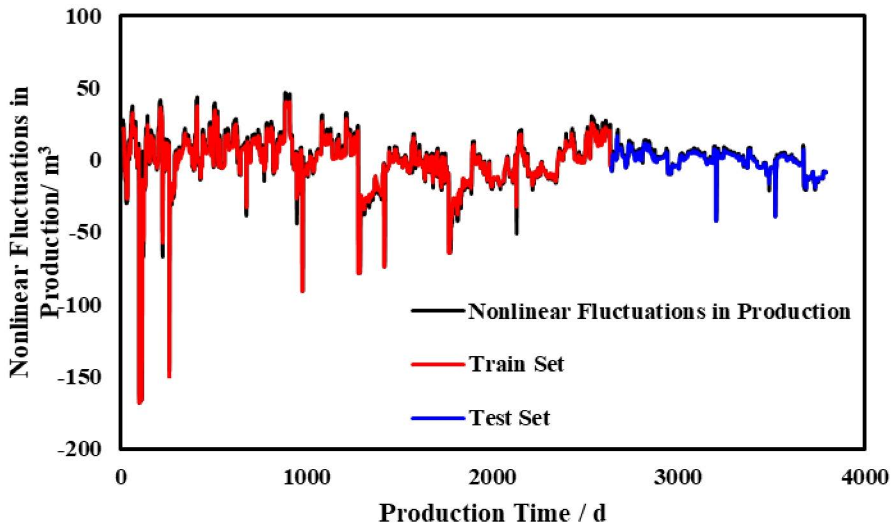

Figure 8—The Production Forecasting Results of Nonlinear Fluctuations in Production

The production forecasting results of Decline Curve Analysis and the Multi-Factor LSTM fusion are as shown in the Fig. 9. This method yields good production forecasting results for this oil well, with an R^2 value of 0.965. This indicates that the method is equally applicable to other oil wells, featuring high accuracy and applicability. Moreover, as mentioned previously, the model recognizes that production is zero during shut-in operations.

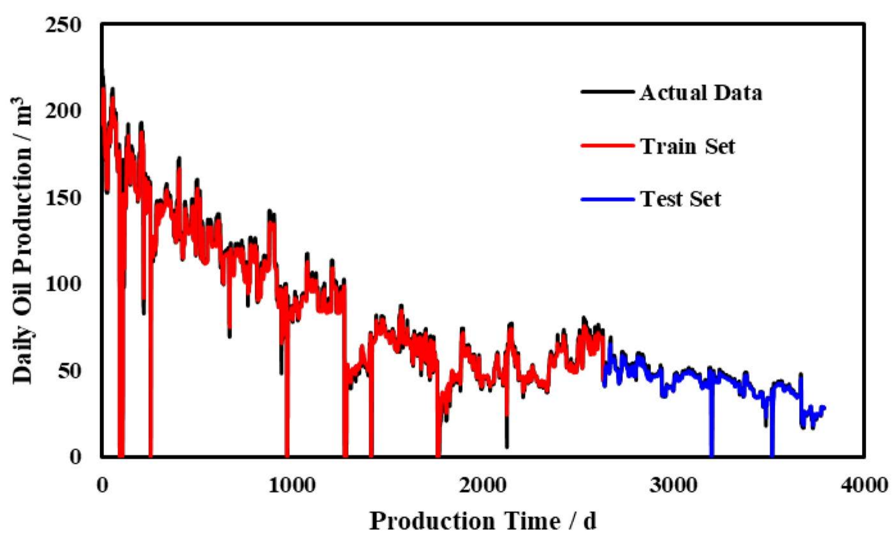

Figure 9—The Production Forecasting Results of Decline Curve Analysis and Multi-Factor LSTM Fusion

Conclusion

To fully leverage the advantages of decline curve analysis and machine learning methods, a production prediction method integrating decline curve analysis and Multi-Factor LSTM was established, achieving a prediction accuracy R^2 of 0.97. This demonstrates that this oil well production prediction method has better prediction capabilities and generalization abilities.

The actual oil well production process is a complex dynamic process influenced by multiple factors. When multiple influencing factors are input into the model, the model can capture the correlations between the data, thereby increasing the accuracy of the oil well production prediction model.

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Nomenclature

LSTM	Long Short-Term Memory
DCA	Decline Curve Analysis
Adam	Adaptive Moment Estimation
GRU	Gated Recurrent Unit

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